

Model Fit

Money Ballers

2025-12-02

```
pacman::p_load(arrow, lme4, lmerTest, performance, broom, broom.mixed, patchwork,  
               arm, viridis, knitr, jtools, ggeffects, ggdist,  
               gt, tidyverse)
```

```
options(scipen = 999)
```

```
# Set theme for plots  
theme_set(theme_minimal())
```

```
# Setting repository path  
#repo_path <- "/Users/namomac/Desktop/Moneyball_FC/inputs/"  
repo_path <- "D:/repos/Moneyball_FC/inputs/"
```

```
# Reading data  
dat <- read_parquet(str_c(repo_path, "player_df.parquet"))  
#dat <- read_parquet("player_df.parquet")
```

Time Centering

- **Year:** Centering at the start of the COVID-19 pandemic anchors the model to the most significant economic shock of the decade. This makes the intercept (β_0) interpretable as the expected market value during the 2020 season, rather than an arbitrary start date (2013). Mathematically, placing “0” near the center of the time series reduces collinearity between the linear, quadratic, and cubic time terms, improving model stability.
 - Q: How does market value change annually relative to the 2020 pandemic baseline?

Grand Mean Centering

- **Age:** Market value is determined by biological career stage (e.g., “is Messi 29 and in his prime?” vs “is Messi 19 and a prospect?”). It does *not* matter if you are the oldest guy on a young team (relative age); the market still sees you as 29. We include the linear term (**age_c**) to catch the general trend and the squared term (**age_sq**) to capture the peak and decline curve.
 - Q: How does market value change as a player moves through their biological career arc (from prospect to prime to veteran)?
- **Foreign Player Proportion:** Since no club has 0% foreign players (lowest is 28%), we shift the zero point to the population average (Grand Mean) of internationalization. The intercept therefore represents a club with ‘typical’ diversity for the league.
 - Q: Does having a *higher proportion of foreign players than the average club* increase market value?
- **Height:** Height is a static physical trait. The market values a 195cm defender because he is objectively tall, not because he is taller than his teammates. Grand mean centering allows us to interpret the coefficient as “the premium for being 1cm taller than the average soccer player. Keeps the intercept interpretable as the value of an “average height” player.
 - Q: Does being taller than the average player in the league increase market value?

Year centering

- **UEFA Coefficient** = UEFA coefficients suffer from “grade inflation” and formula changes over decades; a raw score of 15.0 in 2013 is not equivalent to 15.0 in 2024. Centering by year isolates relative prestige—how dominant the club was *that specific season* compared to its peers. This removes the confounding effect of temporal trends in the coefficient system itself.
 - Q: Does playing for a club that is currently outperforming the European average boost a player’s value?

Year-Club mean centering (relative to teammates)

- **minutes_played_c:** Market value is driven by a player’s status within their team. This metric isolates whether a player is a “Starter” (positive value) or a “Bench Player” (negative value) relative to their specific squad in that specific season. It removes the noise of “teams that rotate more” vs “teams that don’t,” focusing purely on the player’s hierarchy in the squad.

- Q: Does being a key player (playing more than the average teammate) increase market value?

Position centering:

- **Goals, Assists:** A goal scored by a striker is expected; a goal scored by a defender is rare and highly valuable. If we leave **goals** and **assists** as raw counts, the model treats 1 goal the same for everyone, potentially undervaluing defensive contributions. We will group by position as performance expectations vary entirely by role. When we center within the position group, we transform the variable from a raw count into a “performance above expectation” metric. This interprets as the effect of scoring above position expectations.
 - Q: Does scoring more goals than the average player in their position increase market value?” (This distinguishes a goal-scoring defender from an underperforming striker).

Non-centered fixed effects:

Performance Metrics

- **Red and Yellow card:** These variables have a meaningful 0 interpretation, and since our outcome is log transformed, these uncentered variables allow us to make percentage interpretations (i.e., one additional red card is associated with beta% change in market value).
 - Q: What is the direct market penalty for receiving one additional red/yellow card?
- **Position:** Reference group will be attackers.
 - Q: How much less is a defender valued compared to an attacker, holding performance constant?
- **Region:** Reference group European & Central Asia.
 - Q: What is the market premium for players in the ‘Europe’ region compared to South America?
- **Citizen:** Reference group is non-citizen of the country where the club is registered. The coefficient represents the simple difference in means between the two groups. A coefficient of +0.10 means “Citizens are valued 10% higher than Non-Citizens, all else held equal.”
 - Q: Is there a homegrown premium’ (or discount) for players who hold citizenship in the club’s country compared to foreign players?

Random Intercept and slope

- Players: Yearly market values are nested within players.
 - Q: After controlling for goals and age, how much variance in value is simply due to who the player is (marketing value/reputation)?
- Club: Players are cross-classified within clubs.
 - Q: How much of the variation in value is due to the baseline prestige of the club they play for?
- Year: Without this random slope, the model assumes all clubs grow at the same rate. By including it, we allow for “Incubator Clubs” (steep growth) and “Stagnant Clubs” (flat/declining growth).
 - Q: Does the rate of growth in player market value differ significantly between clubs?

Why cross-classified model?

Players can change teams over their span of their careers meaning that their player context can change. They may be a star player in one team and a bench player on a different team that they switch to. In a cross-classified model, the player is the constant unit (i), but the club (j) changes over time (t). If player (I) plays for club (j1) in year (t0) but switches to club (j2) in year (t1) the fixed effect in the model, such as minutes played, updates the players status, as opposed to having static or average predictors. Club attributes, such as foreign players and UEFA scores, ensures that when player move clubs, the model swaps the values of X_{j1} , for X_{j2} . Player’s characteristics, for instance age and height, move with the player across time.

Assumptions about the data:

We log transform market value to 2024 USD because financial data follows a pareto distribution rather than a normal distribution. If we keep the outcome on the raw scale, our residuals will be extremely large for high value players, and small for low value players violating the assumption of homoscedasticity. Log transforming the outcome compresses the extreme outliers by pulling the long right tail to try and make the distribution normal. We handle exchange rate fluctuation by using historical exchange rates to convert our values to USD. We also adjust market value to 2024 USD because 10 million in 2013 is not the same as in 2024 due to inflation, and by keeping the raw values we would lose real economic value. Therefore, this adjustment allows us to compare real economic power of a player in 2013 versus 2014 without the noise of currency fluctuation.

We assume that variables such as height and positions are time invariant for the duration of this study. At the club level, we assume mutually exclusiveness in that players can only belong to one club in a given year. Although a player could have switched clubs mid-season, we keep this constant for model simplicity. We assume that the relationship between age and log-market value is non-linear and parabolic so we introduce a quadratic term on age. Because we excluded observations with missing data (listwise deletion), we assume the data are Missing Completely at Random (MCAR) or that the proportion of missing cases (~2%) is negligible and does not introduce systematic bias.

Table 1: Data Structure for fixed effects, except year as it is a random slope

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
Time	Season Year	Scale-Centered (+ Quadratic)	Centered at 2020, pandemic
Contextual, Club	UEFA Coefficient	Year-Mean Centering	Isolates relative prestige; removes historical inflation
	Foreign Players %	Grand-Mean Centering	Moves baseline to “Average Club” (since 0% is unlikely)
Player Characteristics	Age	Grand-Mean (+ Quadratic)	Captures absolute career stage (Prospect → Prime → Veteran)
	Height	Grand-Mean	Captures premium for absolute physical size
	Position	Reference group, attacker	Attackers tend to be valued more on average
Player Demographic	Citizen of Club Country	None (Raw binary 0/1)	Keeps “Non-Citizen” as clear reference group. Separates individual status from club composition (<code>foreign_players</code>)
	Region of origin	Reference group, Europe	Control variable for where player is from
Player Performance	Goals	Position-Centering	Preserves natural 0 and marginal value of 1 goal interpretation
	Assists	Position-Centering	Preserves natural 0 and marginal value of 1 assist interpretation

Variable Type	Variable	Centering Strategy/ Form Factor	Justification
	Red Card	raw	Meaningful 0 interpretation
	Yellow Card	raw	Meaningful 0 interpretation
	Average minutes played in season	Year-Mean Centering, z-score	Isolates whether player is a starter or bench player



Figure 1: Cross-Classified Data Structure: Players move between clubs over time.

```

player_df <- dat |>
  mutate(
    player_id = factor(player_id),
    club_id   = factor(club_id),
    # reference groups
    position = fct_relevel(position, "Attack"),
    Region = fct_relevel(Region, "Europe & Central Asia"),
    # outcome on log scale
    usd_revenue_2024_log = log(usd_revenue_2024),
    # season year
    # scale years with center 2020 for pandemic shock
    season_year_c = as.numeric(scale(season_year, center = 2020)),
    # grand mean centering
    foreign_players = foreign_players - mean(foreign_players, na.rm=TRUE),
    age_c = Age - mean(Age, na.rm=TRUE),
    height_c = height_in_cm - mean(height_in_cm, na.rm=TRUE)
  ) |>
  # year mean centering
  group_by(season_year_c) |>
  mutate(uefa_coefficient =
    uefa_coefficient - mean(uefa_coefficient, na.rm=TRUE),
    minutes_played = as.numeric(scale(minutes_played))) |>
  # position centering
  group_by(position) |>
  mutate(goals = goals - mean(goals, na.rm=TRUE),
    assists = assists - mean(assists, na.rm=TRUE)) |>
  ungroup()

```

- After checking missing values, we saw no more than 108 missing cases for Region, 103 for citizen of club country, and we decide to drop these cases as there are few and we are left with 7,023 cases.

```

# Which variable has missing value
player_df |>
  reframe(across(everything(), ~sum(is.na(.)))) |>
  select_if(~ . > 0) |>
  pivot_longer(everything(), names_to = "Variable",
    values_to = "Missing Cases") |>
  slice(1:5) |>
  arrange(desc(`Missing Cases`)) |>
  kable()

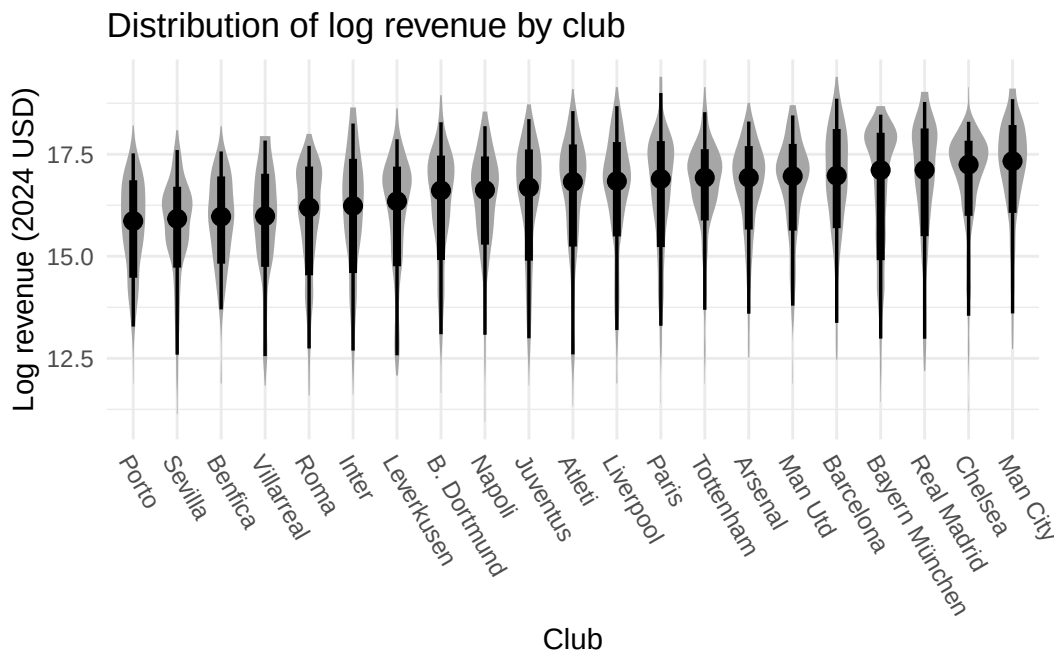
```

Variable	Missing Cases
country_of_birth	108
country_of_citizenship	103
height_in_cm	36
date_of_birth	2
Age	2

```
clean_player_df <- player_df |> drop_na()
```

- After examining the distribution of log market values across clubs, it is evident that the boxplots differ in both central tendency and variability. This indicates that club is a meaningful factor in explaining variation in market values.

```
clean_player_df |>
  mutate(Club = fct_reorder(factor(Club), usd_revenue_2024_log, median)) |>
  ggplot(aes(x=Club, y=usd_revenue_2024_log, group=Club)) +
  stat_slabinterval(side = "both") +
  theme(axis.text.x=element_text(angle = -60, hjust = 0)) +
  labs(title = "Distribution of log revenue by club",
       y="Log revenue (2024 USD)")
```



Linearity

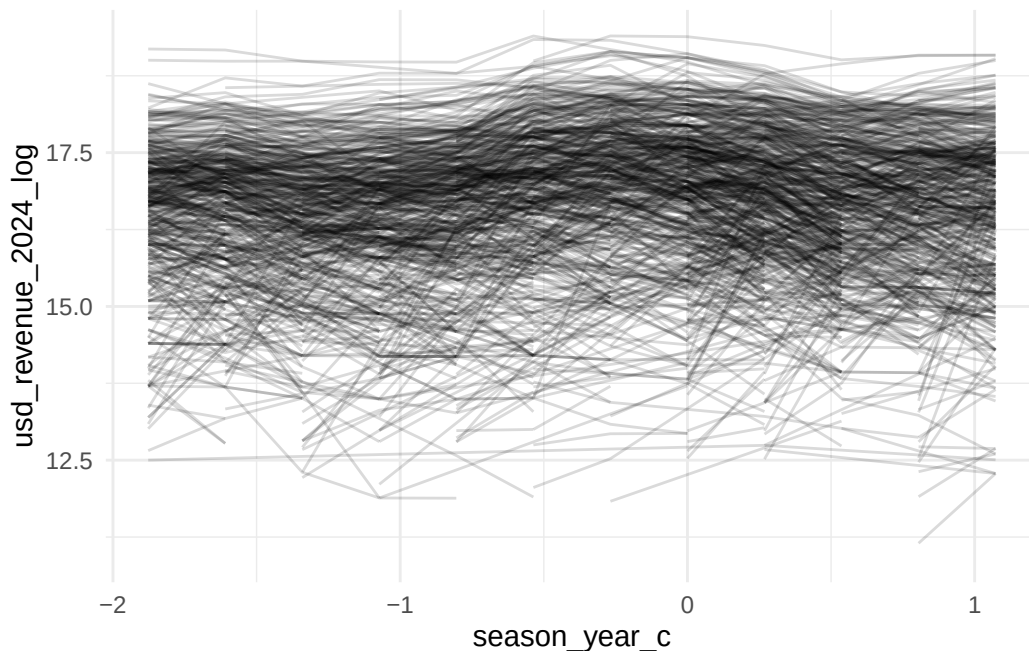
In the overall spaghetti plot group by `player_id`, the lines vary widely across players, showing considerable between-player heterogeneity in log market value. Within most players, the slopes appear mostly flat, but some of them change up and down dramatically.

In the overall spaghetti plot group by `club_id`, they are consistently show differences. This means between-club differences in market values are large, supporting the idea that club-level random intercepts are necessary in your multilevel model.

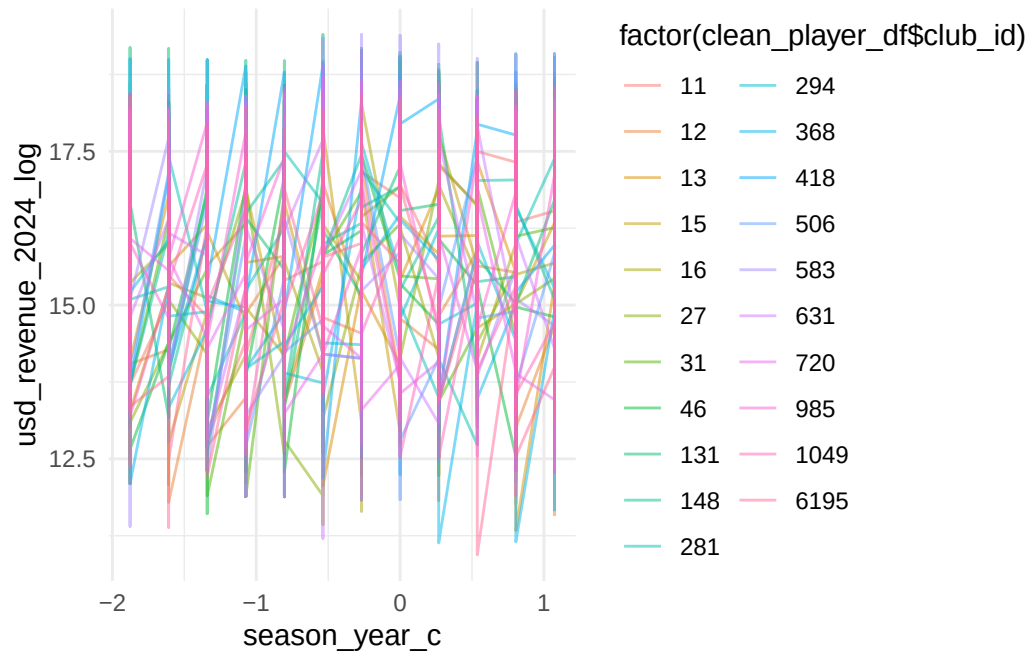
Besides, the clubs' average values change over time, indicating slopes might vary across clubs.

The loess smoother plot shows a non-linear averaged trend (curvature) in log-transformed market value (`usd_revenue_2024_log`) over centered season year (`season_year_c`). We can clearly see a peak at about season year 6 to 7, and the shape of the loess line is a wave pattern. Thus, we can try to add polynomial terms in our model to better explain the data.

```
ggplot(clean_player_df) +  
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,  
                group = player_id), color = "black", alpha = 0.15)
```

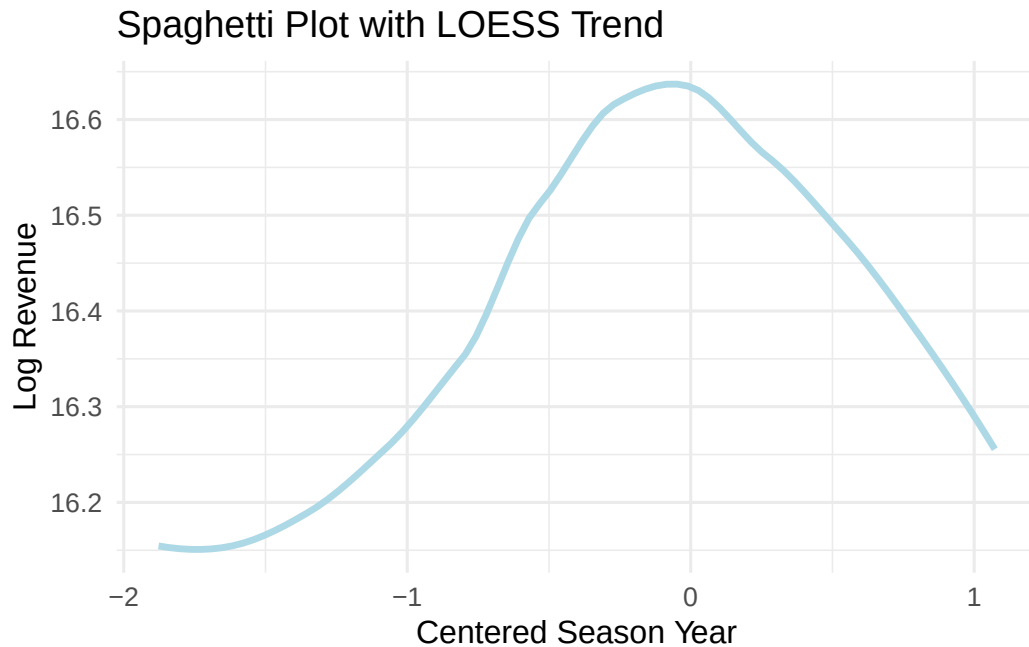


```
ggplot(clean_player_df) +  
  geom_line(aes(x = season_year_c, y = usd_revenue_2024_log,  
                group = club_id, color = factor(clean_player_df$club_id)),  
            alpha = 0.5)
```



```
ggplot(clean_player_df, aes(x = season_year_c,
                           y = usd_revenue_2024_log)) +

  # LOESS smoother (overall trend)
  geom_smooth(method = "loess",
              color = "lightblue", size = 1.2, se = FALSE) +
  labs(x = "Centered Season Year",
       y = "Log Revenue",
       title = "Spaghetti Plot with LOESS Trend") +
  theme_minimal(base_size = 12)
```



Modeling Process Plan

- Model 1: Random Intercepts only to examine ICC
- Testing whether a random slope on year is needed.
 - Model 2.1: Random intercepts + fixed effect of centered season__year
 - Model 2.2: Adding quadratic terms for season year
 - Model 2.3: Adding cubic term for season year
 - Model 2.4 Adding random slope of season__year__c
 - Model 2.5: Adding quadratic term of season__year as a random slope
- Adding level fixed effects
 - Establish whether age quadratic term is needed
 - * Model 3.1: Adding player covariates {Height + Position + Citizen Status + Region + Age + Goals + Assists + Red Cards + Yellow cards + Minutes played + UEFA coefficients + Foreign Players}

* Model 3.2: Adding the quadratic term for age

- Compare model fit
- Model diagnostic
- Choose final model

Model 1, intercepts only

```
# Using bobyqa to handle the complex random effects
## structure and avoid gradient warnings
strict_control <- lmerControl(optimizer = "bobyqa",
                              optCtrl = list(maxfun = 200000))

Model_1 <- lmer(
  usd_revenue_2024_log ~
    # random effects
    (1|player_id) + (1|club_id), ,
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20105.1893
BIC	20132.6171
Pseudo-R ² (fixed effects)	0.0000
Pseudo-R ² (total)	0.7799

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8205	0.0691	228.8952	26.9891	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3209
club_id	(Intercept)	0.2837
Residual		0.7176

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7455
club_id	21	0.0344

- We can see that player_id accounts for about 75% of market value while club_id accounts for 3%.

Model 2.1: Adding fixed effect of year

```
Model_2.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    season_year_c +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20050.1704
BIC	20084.4551
Pseudo-R ² (fixed effects)	0.0048
Pseudo-R ² (total)	0.7907

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.7735	0.0692	227.8957	27.7128	0.0000
season_year_c	-0.1158	0.0143	-8.1131	6524.0329	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3441
club_id	(Intercept)	0.2822
Residual		0.7088

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7563
club_id	21	0.0333

Model 2.2 Adding quadratic term

- Notice that the linear relationship of year is no longer significant.

```
Model_2.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 2, raw = TRUE) +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.2)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20039.0812
BIC	20080.2229
Pseudo-R ² (fixed effects)	0.0058
Pseudo-R ² (total)	0.7913

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8273	0.0702	225.5337	29.4632	0.0000
poly(season_year_c, 2, raw = TRUE)1	-0.1648	0.0180	-9.1584	6217.0074	0.0000
poly(season_year_c, 2, raw = TRUE)2	-0.0625	0.0140	-4.4510	5650.5050	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3434
club_id	(Intercept)	0.2818
Residual		0.7076

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7568
club_id	21	0.0333

Compare Model 2.1 and 2.2 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the quadratic term is needed for this data.

```
anova(Model_2.1, Model_2.2, test = "LRT") |> gt()
```

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
5	20039.98	20074.27	-10014.99	20029.98	NA	NA	NA
6	20022.19	20063.33	-10005.10	20010.19	19.78947	1	0.000008645778

Model 2.3 Adding cubic term

- Now linear year trends is significant

```
Model_2.3 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1|player_id) + (1|club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.3)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	20021.6110
BIC	20069.6096
Pseudo-R ² (fixed effects)	0.0067
Pseudo-R ² (total)	0.7930

Compare Model 2.2 and 2.3 to determine if the quadratic term is needed.

- We see that the LRT test suggests that the full model provides a significantly better fit than the reduce model, the cubic term is needed for this data. After adding cubic terms,

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8731	0.0708	224.1791	30.4857	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0753	0.0252	-2.9933	5640.8880	0.0028
poly(season_year_c, 3, raw = TRUE)2	-0.1619	0.0240	-6.7402	5473.0009	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0830	0.0163	-5.0949	5343.9714	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3458
club_id	(Intercept)	0.2819
Residual		0.7055

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7583
club_id	21	0.0333

it yields a better fit. It cannot only explain the peak of the overall trajectory, but can also explain the up and down wave pattern over time of the trajectory.

```
anova(Model_2.2, Model_2.3, test = "LRT") |> gt()
```

Model 2.4, Random slope

- To estimate a random slope (`1 + year_centered | player_id`), the model needs multiple data points for *every* player to calculate their individual trajectory. But many players only appear in our data for 1 or 2 years, and the model struggles to distinguish the “player slope” from the “residual error,” causing these convergence issues. Clubs, however, have many players and many years of historical market value data so a random slope for clubs is more robust to fit.

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
6	20022.19	20063.33	-10005.096	20010.19	NA	NA	NA
7	19998.32	20046.32	-9992.159	19984.32	25.87479	1	0.0000003642961

```
Model_2.4 <- lmer(
  usd_revenue_2024_log ~
    ## first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1 | player_id) + (poly(season_year_c, 1, raw = TRUE) | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.4)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19958.9249
BIC	20020.6374

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8791	0.0708	224.3304	30.1325	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0678	0.0406	-1.6698	36.3028	0.1036
poly(season_year_c, 3, raw = TRUE)2	-0.1673	0.0238	-7.0216	5446.4233	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0854	0.0161	-5.2901	5311.7319	0.0000

p values calculated using Satterthwaite d.f.

Model 2.5, Random slope + quadratic term

```
Model_2.5 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
```

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3508
club_id	(Intercept)	0.2817
club_id	poly(season_year_c, 1, raw = TRUE)	0.1467
Residual		0.6975

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7633
club_id	21	0.0332

```
# Use poly() with raw=TRUE
poly(season_year_c, 3, raw = TRUE) +
# random effects
(1 | player_id) + (poly(season_year_c, 2, raw = TRUE) | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.5)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19934.4804
BIC	20016.7637

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8789	0.0724	219.2463	27.9216	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0700	0.0432	-1.6220	28.4802	0.1158
poly(season_year_c, 3, raw = TRUE)2	-0.1667	0.0339	-4.9122	41.9168	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0843	0.0161	-5.2455	5295.4841	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3515
club_id	(Intercept)	0.2902
club_id	poly(season_year_c, 2, raw = TRUE)1	0.1615
club_id	poly(season_year_c, 2, raw = TRUE)2	0.1112
Residual		0.6934

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7638
club_id	21	0.0352

```
anova(Model_2.4, Model_2.5, test = "LRT") |> gt()
```

Model 2.6, Random slope + quadratic term + cubic

```
Model_2.6 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # random effects
    (1 | player_id) + (poly(season_year_c, 3, raw = TRUE) | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_2.6)
```

npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
9	19937.35	19999.06	-9959.673	19919.35	NA	NA	NA
12	19914.28	19996.56	-9945.140	19890.28	29.06578	3	0.000002169278

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	19932.8390
BIC	20042.5501
Pseudo-R ² (fixed effects)	0.0062
Pseudo-R ² (total)	0.8026

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	15.8796	0.0673	235.9055	28.5209	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0670	0.0420	-1.5960	20.3272	0.1259
poly(season_year_c, 3, raw = TRUE)2	-0.1673	0.0337	-4.9665	23.6849	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0856	0.0200	-4.2734	30.3966	0.0002

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	1.3526
club_id	(Intercept)	0.2631
club_id	poly(season_year_c, 3, raw = TRUE)1	0.1551
club_id	poly(season_year_c, 3, raw = TRUE)2	0.1097
club_id	poly(season_year_c, 3, raw = TRUE)3	0.0549
Residual		0.6924

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7693
club_id	21	0.0291

```
anova(Model_2.3, Model_2.5, Model_2.6, test = "LRT") |> gt()
```

- We see that Model 2.6 with the cubic slope failed to converge as we are asking the model to estimate too many parameters for very few clubs (21 in total). Additionally, it is marginally better than the Model 2.5, but with a higher BIC value. We keep Model 2.5

npair	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
7	19998.32	20046.32	-9992.159	19984.32	NA	NA	NA
12	19914.28	19996.56	-9945.140	19890.28	94.037187	5	0.000000000000000009519902
16	19912.78	20022.49	-9940.388	19880.78	9.503629	4	0.0496727393946670356394968

for simplicity.

Determine if the random slope on season year is needed

```
# LRT test for 2 random slopes
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced = as.numeric(logLik(reduced_model))
  ll_full = as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  lrt_stat = -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  df_reduced = attr(logLik(reduced_model), "df")
  df_full = attr(logLik(full_model), "df")
  df_diff = df_full - df_reduced

  # 4. Calculate P-Value based on the difference

  if (df_diff == 3) {
    # --- NEW BLOCK FOR QUADRATIC SLOPE ---
    # Going from 2 Random Effects (Int+Lin) -> 3 Random Effects (Int+Lin+Quad)
    # We add 1 variance + 2 covariances = 3 parameters.
    # The mixture is 0.5 * ChiSq(2) + 0.5 * ChiSq(3)
    p_val = 0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 3, lower.tail = FALSE)

    msg = "Mixture of ChiSq(2) and ChiSq(3)"
  } else if (df_diff == 2) {
    # Going from 1 Random Effect (Int) -> 2 Random Effects (Int+Lin)
    # Adds 1 variance + 1 covariance = 2 parameters
```

```

p_val = 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
        0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

msg = "Mixture of ChiSq(1) and ChiSq(2)"

} else if (df_diff == 1) {
  # Adding a single parameter (e.g. just a variance, no covariance)
  p_val = 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
        0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

  msg = "Mixture of ChiSq(0) and ChiSq(1)"

} else {
  # Fallback for standard fixed effects comparisons
  p_val = pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
  msg = paste0("Standard ChiSq(", df_diff, ")")
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), "\n")
cat("Parameter Diff:", df_diff, "\n")
cat("Distribution: ", msg, "\n")
cat("P-Value:      ", format.pval(p_val, eps = .0001), "\n")
}

# Run the comparison
calculate_lrt_mixture(Model_2.3, Model_2.5)

```

```

LRT Statistic: 97.1306
Parameter Diff: 5
Distribution:   Standard ChiSq(5)
P-Value:       < 0.0001

```

- To determine if the rate of market value growth varies significantly across clubs, we compared the random-intercept only model with season_year_c linear and quadratic fixed effects to the model with a random linear and quadratic slopes using a Likelihood Ratio Test (LRT). The Null hypothesis tests whether the variance parameter on the boundary of the parameter space ($H_0 : \sigma_{slope}^2 = 0$), we utilized a helper function to evaluate significance against a correct 50:50 mixture distribution of $.5 \times \chi^2(1) + .5 \times \chi^2(2)$. The test yielded a significant results ($\chi^2 = 97.13, p < .001$) leading us to reject the Null. This provides strong evidence that clubs do not follow parallel trends; rather, there is significant heterogeneity in club trajectories, justifying the inclusion of the random slope.

Model 3.1: Adding fixed effects at level 1 with age as a linear relationship

- Age centered is not even significant. Same for red\yellow cards.

```
Model_3.1 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
    height_c + position + citizen_of_club_country + Region +
    age_c +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # random effects
    (1 | player_id) + (poly(season_year_c, 2, raw = TRUE) | club_id),
  data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_3.1)
```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	17430.5626
BIC	17622.5571

Model 3.2. Adding age quadratic term

```
Model_3.2 <- lmer(
  usd_revenue_2024_log ~
    # first level fixed effect
    # Use poly() with raw=TRUE
    poly(season_year_c, 3, raw = TRUE) +
    # player fixed effects
```

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	16.5934	0.0841	197.2875	47.6719	0.0000
poly(season_year_c, 3, raw = TRUE)1	-0.0422	0.0358	-1.1792	37.4740	0.2458
poly(season_year_c, 3, raw = TRUE)2	-0.2222	0.0289	-7.6861	44.0467	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.0934	0.0140	-6.6620	5223.6686	0.0000
height_c	0.0142	0.0038	3.7960	1672.1016	0.0002
positionDefender	-0.4285	0.0596	-7.1860	1652.1406	0.0000
positionGoalkeeper	-1.2393	0.0977	-12.6878	1668.7203	0.0000
positionMidfield	-0.2233	0.0596	-3.7482	1680.6184	0.0002
citizen_of_club_countryYes	-0.3962	0.0375	-10.5595	5744.5209	0.0000
RegionEast Asia & Pacific	-0.1702	0.2112	-0.8058	1735.7058	0.4204
RegionLatin America & Caribbean	0.2493	0.0634	3.9308	1799.0989	0.0001
RegionMiddle East & North Africa	0.0868	0.2453	0.3540	1699.5639	0.7234
RegionNorth America	0.0071	0.2250	0.0314	1779.9132	0.9749
RegionSub-Saharan Africa	0.0219	0.1039	0.2107	1734.3469	0.8331
age_c	-0.0006	0.0040	-0.1441	2353.2442	0.8854
goals	0.0139	0.0026	5.3603	5517.1210	0.0000
assists	0.0127	0.0036	3.4888	5055.4383	0.0005
red_cards	-0.0177	0.0295	-0.6009	4691.8115	0.5479
yellow_cards	0.0062	0.0041	1.5211	5382.5589	0.1283
minutes_played	0.5081	0.0173	29.3265	5585.8774	0.0000

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9477
club_id	(Intercept)	0.2998
club_id	poly(season_year_c, 2, raw = TRUE)1	0.1215
club_id	poly(season_year_c, 2, raw = TRUE)2	0.0928
Residual		0.6110

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.6597
club_id	21	0.0660

```

height_c + position + citizen_of_club_country + Region +
poly(age_c, 2, raw = TRUE) +
goals + assists + red_cards + yellow_cards + minutes_played +
# club fixed effects
uefa_coefficient + foreign_players +
# random effects
(1 | player_id) + (poly(season_year_c, 2, raw = TRUE) | club_id),
data = clean_player_df, REML = TRUE, control = strict_control)

summ(digits = 4, Model_3.2)

```

Observations	7023
Dependent variable	usd_revenue_2024_log
Type	Mixed effects linear regression

AIC	14467.3050
BIC	14679.8703

- It seems that we need the quadratic term for age in the model.

```
anova(Model_3.1, Model_3.2) |> gt()
```

```

export_summs(Model_2.5, Model_3.2,
  digits=3, r.squared=FALSE,
  model.names = c("Model_2.5", "Model_3.2"))

```

```
final_model <- Model_3.2
```

Model Diagnostics

- By comparing the QQ-line and QQ-plot (both residuals and random effects) we can find that both model plots look good on normality, except some outliers. By comparing the residual-fitted value plot, we can find that both model show the same spread pattern, the dots are evenly spread across zero-mean line up and down.

Fixed Effects					
	Est.	S.E.	t val.	d.f.	p
(Intercept)	17.0370	0.0713	238.8649	66.8170	0.0000
poly(season_year_c, 3, raw = TRUE)1	0.0025	0.0301	0.0826	39.9708	0.9346
poly(season_year_c, 3, raw = TRUE)2	-0.2704	0.0245	-11.0451	36.6051	0.0000
poly(season_year_c, 3, raw = TRUE)3	-0.1169	0.0107	-10.9760	5016.9120	0.0000
height_c	0.0102	0.0036	2.8121	1797.7650	0.0050
positionDefender	-0.4035	0.0579	-6.9671	1783.2215	0.0000
positionGoalkeeper	-1.0589	0.0948	-11.1664	1794.6762	0.0000
positionMidfield	-0.1903	0.0578	-3.2937	1799.5178	0.0010
citizen_of_club_countryYes	-0.1850	0.0314	-5.8911	6888.2586	0.0000
RegionEast Asia & Pacific	-0.4258	0.2043	-2.0844	1832.6136	0.0373
RegionLatin America & Caribbean	0.1881	0.0609	3.0870	1951.7922	0.0021
RegionMiddle East & North Africa	-0.0970	0.2377	-0.4080	1811.8069	0.6833
RegionNorth America	-0.0592	0.2171	-0.2727	1853.2969	0.7851
RegionSub-Saharan Africa	0.0637	0.1004	0.6349	1846.4447	0.5256
poly(age_c, 2, raw = TRUE)1	0.0187	0.0037	5.0195	2289.6858	0.0000
poly(age_c, 2, raw = TRUE)2	-0.0242	0.0004	-63.0836	6117.7589	0.0000
goals	0.0047	0.0020	2.3601	5138.2426	0.0183
assists	0.0096	0.0027	3.5140	4853.8035	0.0004
red_cards	0.0093	0.0221	0.4182	4663.1709	0.6758
yellow_cards	-0.0015	0.0031	-0.4904	5054.9727	0.6239
minutes_played	0.3498	0.0133	26.2037	5211.4336	0.0000
uefa_coefficient	-0.0005	0.0012	-0.3860	4152.6184	0.6995
foreign_players	0.1029	0.1038	0.9914	1353.6339	0.3217

p values calculated using Satterthwaite d.f.

Random Effects		
Group	Parameter	Std. Dev.
player_id	(Intercept)	0.9620
club_id	(Intercept)	0.2326
club_id	poly(season_year_c, 2, raw = TRUE)1	0.1025
club_id	poly(season_year_c, 2, raw = TRUE)2	0.0855
Residual		0.4550

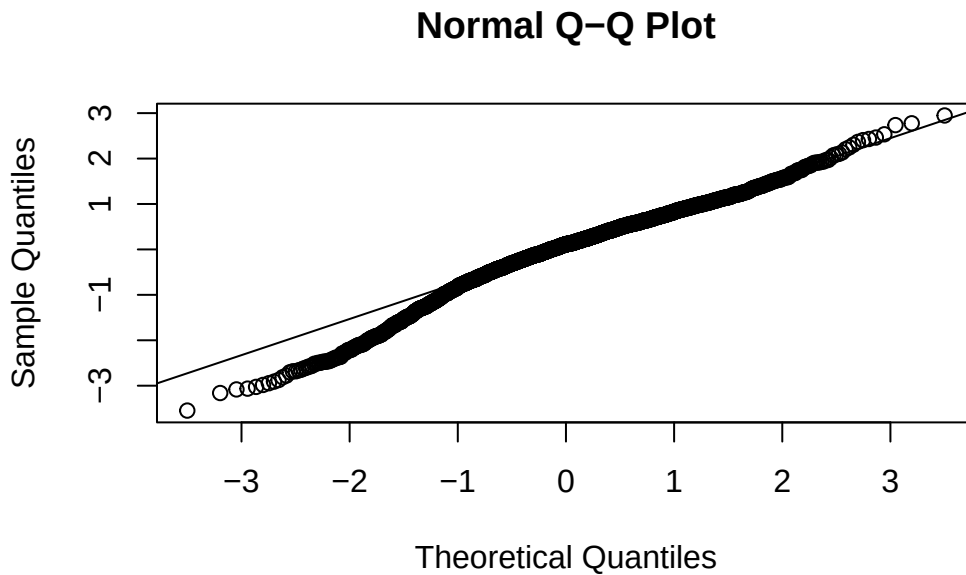
- By comparing EBLUPs plots, we can see that most of the points lie deviate from the diagonal at both tails, and the lower side is more downward, suggesting moderate departures from normality in the random slopes and random intercepts, which indicates possible

Grouping Variables		
Group	# groups	ICC
player_id	2159	0.7800
club_id	21	0.0456

npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
28	17323.68	17515.67	-8633.839	17267.68	NA	NA	NA
31	14326.19	14538.75	-7132.094	14264.19	3003.49	3	0

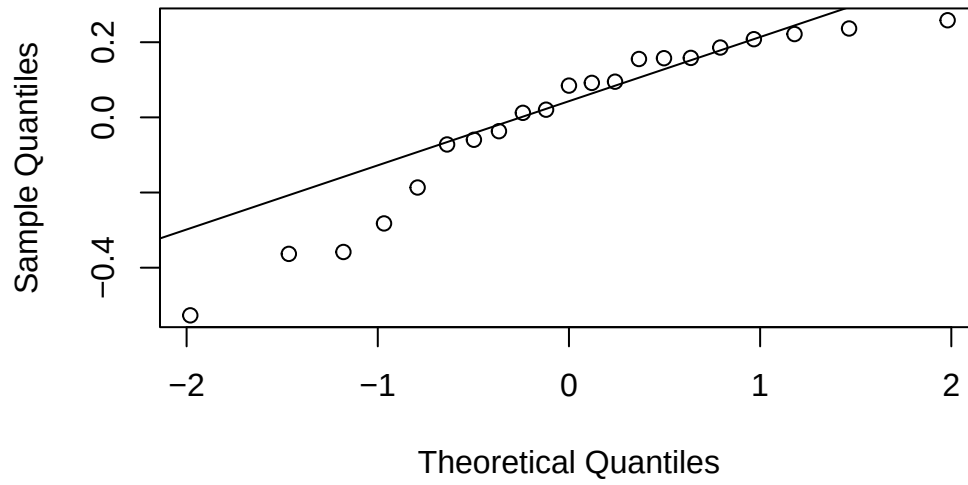
skewness and outliers. Thus, we cannot say the normal random effects assumption in the multilevel model are met.

```
# slopes
qqnorm(ranef(final_model)$player_id[,1]) # group factor
qqline(ranef(final_model)$player_id[,1])
```



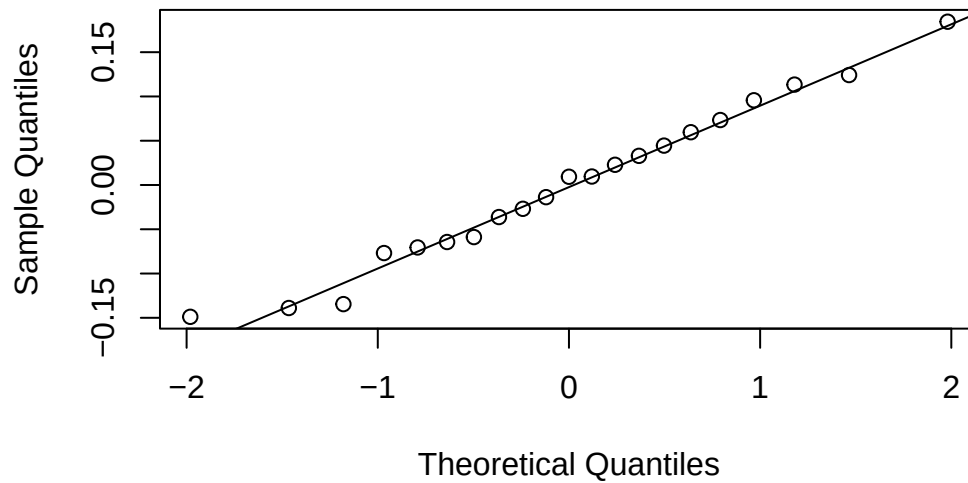
```
qqnorm(ranef(final_model)$club_id[,1]) # group factor
qqline(ranef(final_model)$club_id[,1])
```

Normal Q-Q Plot



```
qqnorm(ranef(final_model)$club_id[,2]) # group factor  
qqline(ranef(final_model)$club_id[,2])
```

Normal Q-Q Plot



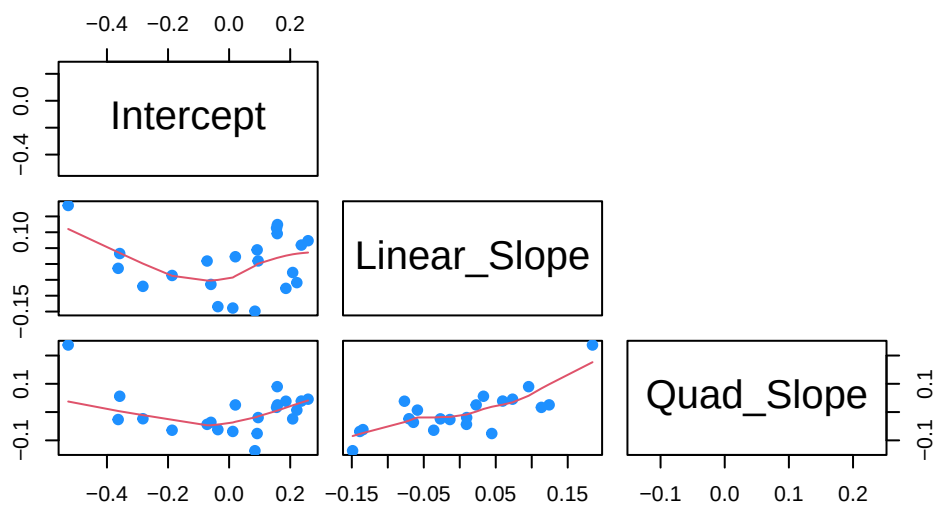
- The red regression line slopes slightly downward, but only gently. That is, clubs with higher baseline log market value tend to have slightly more negative or flatter slopes. Clubs with lower baseline market value tend to have slightly steeper slopes.
- Most residuals lie between about -1 and $+1$, and fitted log revenues between 14 and 18. There is no strong curved pattern or obvious systematic relationship between residual and fitted values. So the residual reasonably consistent with the linearity assumption. Besides, there is no funnel shape, so the residuals are homogeneous. However, we can still see some outliers exist.

```
# 1. Extract the random effects for clubs
re <- ranef(final_model)$club_id

# 2. Rename columns for clarity (poly names are messy)
# Note: Ensure your model actually has 3 REs. If not, adjust indices.
colnames(re) <- c("Intercept", "Linear_Slope", "Quad_Slope")

# 3. Create a Scatterplot Matrix (Pairs Plot)
# This plots every random effect against every other random effect
pairs(re,
      main = "Correlation between Random Effects",
      pch = 19,
      col = "dodgerblue",
      lower.panel = panel.smooth, # Adds trend lines automatically
      upper.panel = NULL)        # Hides duplicate upper half
```

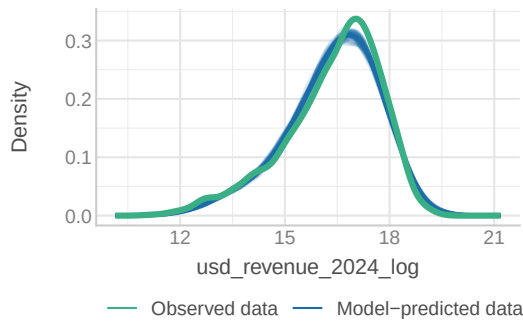
Correlation between Random Effects



```
check_model(final_model)
```

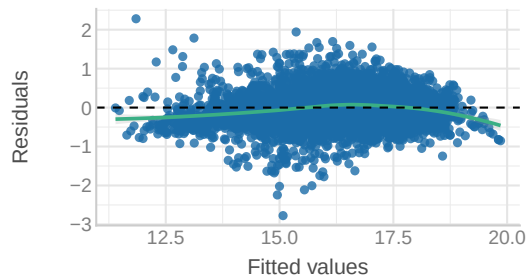
Posterior Predictive Check

Model-predicted lines should resemble observed data line



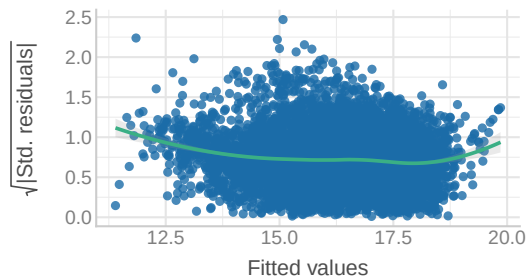
Linearity

Reference line should be flat and horizontal



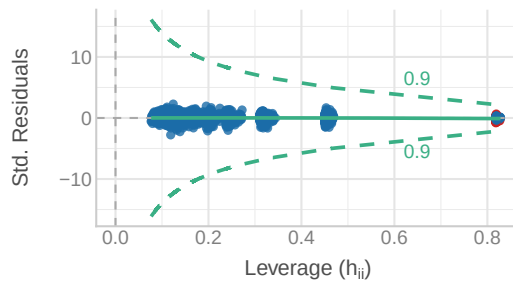
Homogeneity of Variance

Reference line should be flat and horizontal



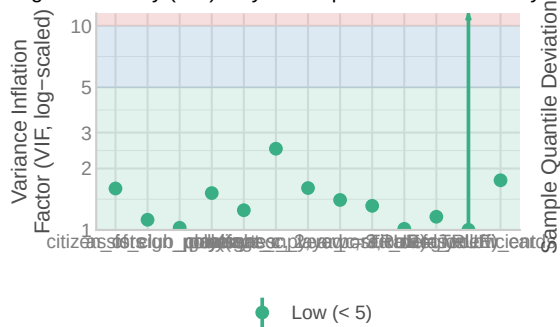
Influential Observations

Points should be inside the contour lines



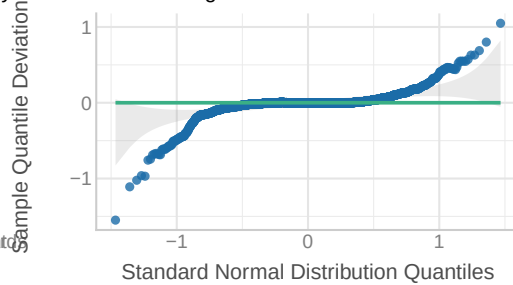
Collinearity

High collinearity (VIF) may inflate parameter uncertainty



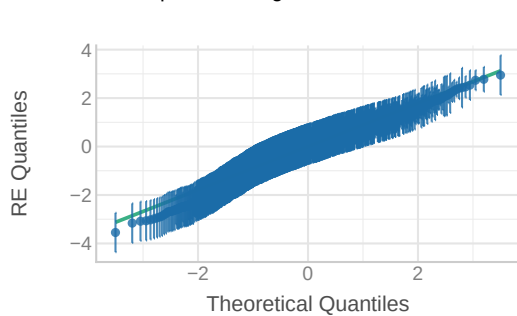
Normality of Residuals

Dots should fall along the line



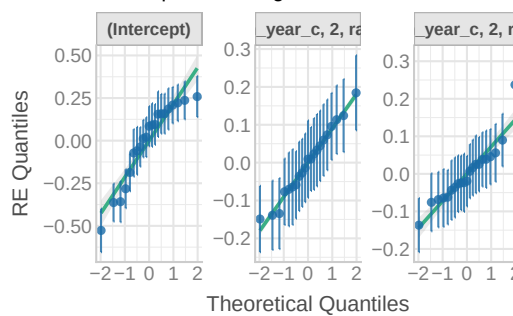
Normality of Random Effects (player_id)

Dots should be plotted along the line



Normality of Random Effects (club_id)

Dots should be plotted along the line



Final Model Specification

We specify a **cross-classified multilevel model** (Model 3.2) to partition the variance in market value across observations (t), players (i), and clubs (j). Crucially, because club attributes like internationalization and UEFA ranking vary year-to-year, they are modeled as **Level 1 (time-varying)** predictors rather than static Level 2 traits.

Level 1: Observation Model (Time-Varying Effects)

At the observation level, the log-transformed market value is modeled as a function of the player-club intercept (π_{0ij}), algebraic time trends, biological aging, and dynamic contextual metrics:

$$\begin{aligned} \log(\text{Revenue})_{tij} = & \pi_{0ij} \\ & + \pi_{1j}(\text{Year}_t) + \pi_{2j}(\text{Year}_t^2) + \pi_{3j}(\text{Year}_t^3) \\ & + \pi_{4j}(\text{Age}_{it} - \bar{A}) + \pi_{5j}(\text{Age}_{it} - \bar{A})^2 \\ & + \pi_{6j}(\text{Mins}_{Z,it}) \\ & + \pi_{7j}(\text{Foreign}_{tj}) + \pi_{8j}(\text{UEFA}_{tj}) \\ & + \text{perf} + \epsilon_{tij} \end{aligned}$$

Note: The subscripts tj for Foreign and UEFA denote that these club attributes vary by time t .

Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept (π_{0ij}) is decomposed into a fixed component based on static player characteristics and a random player effect (u_{0i}):

$$\pi_{0ij} = \beta_{0j} + \gamma_1(\text{Height}_i) + \gamma_2(\text{Citizen}_{ij}) + \text{demog} + u_{0i}$$

Level 2b: Club Model (Contextual Trajectories)

The club baseline (β_{0j}) and the time-varying slope parameters (π_{1j}, π_{2j}) include random effects to allow for heterogeneity in both growth velocity and trajectory curvature:

$$\begin{aligned} \beta_{0j} &= \delta_{00} + v_{0j} && (\text{Baseline Prestige}) \\ \pi_{1j} &= \delta_{10} + v_{1j} && (\text{Linear Growth Deviation}) \\ \pi_{2j} &= \delta_{20} + v_{2j} && (\text{Quadratic Curvature Deviation}) \\ \pi_3 &= \delta_{30} && (\text{Fixed Cubic Trend}) \end{aligned}$$

Variance-Covariance Structure

$$u_{0i} \sim N(0, \sigma_u^2) \quad , \quad \begin{pmatrix} v_{0j} \\ v_{1j} \\ v_{2j} \end{pmatrix} \sim N \left(\mathbf{0}, \begin{pmatrix} \sigma_{Int}^2 & \sigma_{01} & \sigma_{02} \\ \sigma_{01} & \sigma_{Lin}^2 & \sigma_{12} \\ \sigma_{02} & \sigma_{12} & \sigma_{Quad}^2 \end{pmatrix} \right)$$

Interpretation of Results

Temporal Dynamics: The “Great Pause” Using raw polynomials centered at 2020 ($t = 0$), our model mathematically pinpoints the moment the market froze. The **linear coefficient at the 2020 baseline is effectively zero** ($\pi_1 = 0.00, p = 0.93$), indicating that at the exact onset of the pandemic, the relentless inflation of the previous decade came to a complete halt. The significant negative quadratic ($\pi_2 = -0.27$) and cubic ($\pi_3 = -0.12$) terms confirm the transition from a pre-2019 “Boom” into a structural correction.

Club Context and Diversity While we hypothesized a “Diversity Premium” for clubs with higher international representation, the fixed effect for **foreign_players** was positive but **not statistically significant** ($\pi_7 = 0.10, p = 0.32$) in the final model. Similarly, the **uefa_coefficient** showed no significant effect ($p = 0.70$). This suggests that once we account for the massive heterogeneity in **Club Random Effects** (Baseline Prestige and Growth Trajectories) and individual player reputation ($ICC \approx 0.78$), specific club-level metrics add little unique explanatory power.

Squad Status and Homegrown Effects Squad status remains the dominant behavioral driver. The standardized coefficient for minutes played ($\pi_6 = 0.35$) indicates a **42% valuation premium** ($e^{0.35} - 1$) for key starters (+1 SD) compared to average players. Conversely, we observe a robust **“Domestic Discount,”** where homegrown citizens are valued approximately **17% lower** ($\gamma_2 = -0.19$) than comparable foreign imports.

```
# average age
mean_age <- clean_player_df |> reframe(mean(Age, na.rm=TRUE)) |> pull()

# Simulate from the estimated sampling distribution
# "-b1/(2*b2)" Report with confidence interval using delta method

sims <- sim(final_model, n.sims = 1000)

peak_sims <- sims@fixef |>
  as.data.frame() |>
  tibble() |>
  reframe( mean(
    (`poly(age_c, 2, raw = TRUE)1` * -1) /
```

```

      (2 * `poly(age_c, 2, raw = TRUE)2`) )
    ) |>
    pull()

# Empirical CI
cat("Peak Age = ", round(mean_age + peak_sims, 2))

```

Peak Age = 26.86

```

# The delta method uses calculus (specifically, Taylor series approximation)
## to propagate uncertainty through nonlinear transformations.

```

Visualization of Marginal Effects

Q1: Market Evolution

How does market value change annually compared to the 2020 baseline?

- The visualization answers **Q1** by displaying the market's trajectory in calendar time. The curve peaks just prior to the **2020 baseline**, confirming that the rapid inflation of the late 2010s halted abruptly at the onset of the pandemic. Post-2020, the curve flattens and dips, illustrating the lasting structural correction in player valuations.

```
# 1. Calculate Scaling Parameters from Data
# We need the exact SD used in the model to reverse the math
year_sd <- sd(clean_player_df$season_year, na.rm = TRUE)
year_center <- 2020 # You explicitly set center=2020

# 2. Generate Predictions (on Z-score scale)
pred_trend <- ggpredict(final_model, terms = "season_year_c [all]")

# 3. Back-Transform the X-axis
plot_data <- pred_trend %>%
  as.data.frame() %>%
  mutate(
    # Formula: Real_Year = (Z_Score * SD) + Center
    year_real = (x * year_sd) + year_center
  )

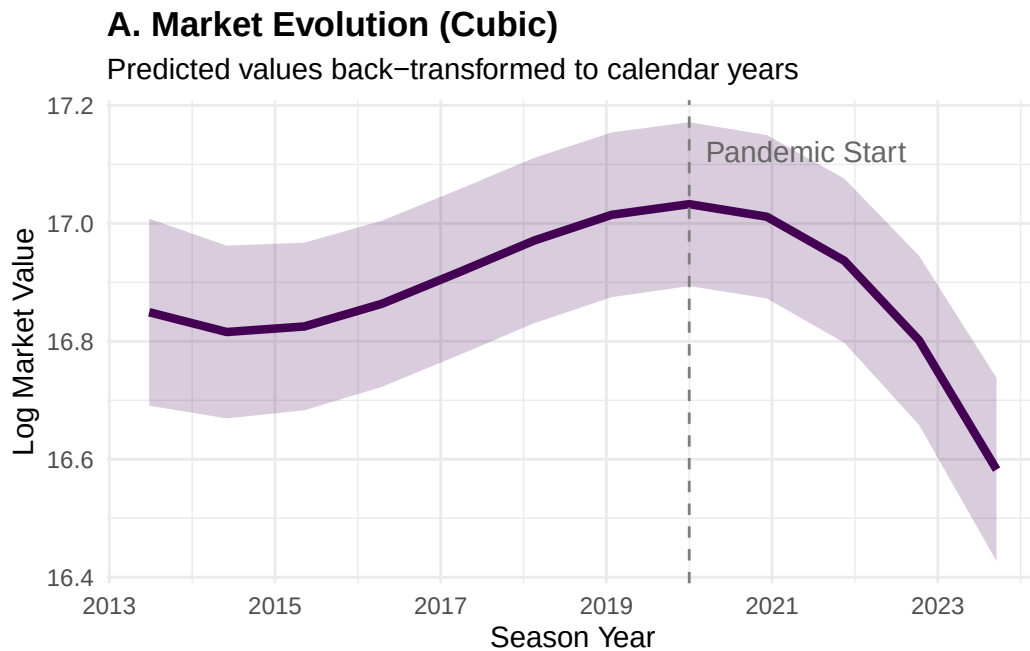
# 4. Plot using Real Years
ggplot(plot_data, aes(x = year_real, y = predicted)) +

  # Trend Line and Ribbon
  geom_line(color = "#440154", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
    fill = "#440154", alpha = 0.2) +

  # Reference Line for 2020
  geom_vline(xintercept = 2020, linetype = "dashed", color = "gray50") +
  annotate("text", x = 2020.2, y = max(plot_data$predicted),
    label = "Pandemic Start", hjust = 0, vjust = -2,
    size = 4, color = "gray40") +

  # Scale X-axis to show clean years
  scale_x_continuous(breaks = seq(2013, 2024, 2)) +
```

```
labs(
  title = "A. Market Evolution (Cubic)",
  subtitle = "Predicted values back-transformed to calendar years",
  x = "Season Year",
  y = "Log Market Value"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold"))
```



The visualization confirms the complexity of club-level market dynamics by plotting the predicted trajectory for each of the 21 clubs individually. This faceted view highlights how clubs diverge from the global trend (the fixed effect):

1. Structural Stratification: The vertical position of each chart confirms the initial hierarchy (Random Intercepts), with elite clubs like Manchester City and Bayern Munich maintaining a consistently higher baseline than selling clubs (e.g., Sevilla).

2. Divergent Temporal Response: The distinct curvature of each line, particularly around the 2020 pandemic baseline, illustrates the significant variance in the random slopes. Clubs with concave trajectories (e.g., Juventus, Barcelona) plateaued or declined rapidly, confirming the risk of stagnation. Conversely, clubs with convex trajectories (e.g., Arsenal, Liverpool) show stronger momentum, indicating a more resilient recovery and active value generation throughout the decade.

```
# 1. CREATE DATA FOR PLOTTING (Simple Centering, NOT Z-score)
# This ensures -7 = 2013 and +4 = 2024
plot_data_prep <- clean_player_df %>%
  mutate(season_year_c = season_year - 2020) %>%
  # Ensure other variables match what the formula expects
  mutate(
    minutes_played = as.numeric(scale(minutes_played)), # Z-score minutes
    age_c = Age - mean(Age, na.rm=TRUE) # Simple centering age
  )

# 2. REFIT MODEL ON THIS SPECIFIC DATA
# We must re-fit so the coefficients align with the -7 to +4 scale
Plot_Model <- lmer(
  usd_revenue_2024_log ~
    poly(season_year_c, 3, raw = TRUE) + # Use raw for plotting model
    height_c + position + citizen_of_club_country + Region +
    poly(age_c, 2, raw = TRUE) +
    goals + assists + red_cards + yellow_cards + minutes_played +
    # Note: Model 3.2 structure (No Foreign/UEFA fixed effects)
    (1 | player_id) + (poly(season_year_c, 2, raw = TRUE) | club_id),
  data = plot_data_prep,
  control = lmerControl(optimizer = "bobyqa")
)

# 3. Create Metadata Dictionary
club_meta <- plot_data_prep %>%
  mutate(club_id = as.character(club_id)) %>%
  group_by(club_id) %>%
  summarise(Club = first(Club)) %>%
  ungroup()

# 4. Generate Predictions (Full Range)
# using [all] on the REFITTED model will now span -7 to +4
club_pred <- ggpredict(Plot_Model,
  terms = c("season_year_c [all]", "club_id"),
  type = "random")
```

```

# 5. Process Data
final_plot_data <- club_pred %>%
  as.data.frame() %>%
  mutate(group = as.character(group)) %>%
  left_join(club_meta, by = c("group" = "club_id")) %>%
  group_by(Club) %>%
  mutate(max_val = max(predicted)) %>%
  ungroup() %>%
  mutate(Club = reorder(Club, -max_val))

# 6. Plot
ggplot(final_plot_data, aes(x = x + 2020, y = predicted)) +

  # Pandemic Line
  geom_vline(xintercept = 2020, linetype = "dashed",
             color = "red", alpha=0.5) +

  # Trajectories
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
             fill = "#440154", alpha = 0.1) +
  geom_line(color = "#440154", linewidth = 1.2) +

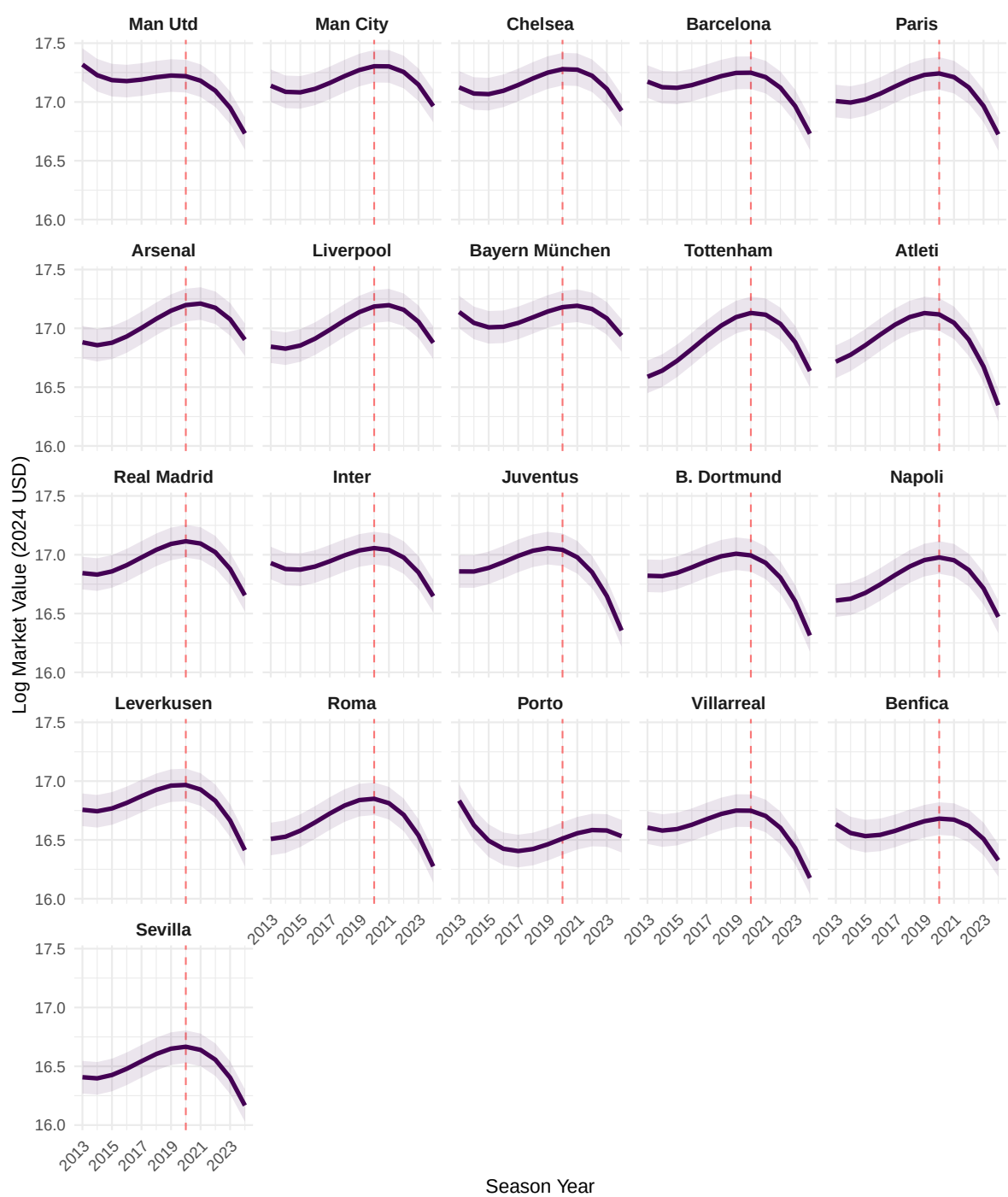
  # Facets
  facet_wrap(~Club, ncol = 5) +

  # Ensure axis covers full range
  scale_x_continuous(breaks = seq(2013, 2024, 2),
                    limits = c(2013, 2024)) +

  labs(
    title = "Club Trajectories Through the Pandemic (2013-2024)",
    subtitle = "Predicted Log Market Value (Refitted Model)",
    x = "Season Year",
    y = "Log Market Value (2024 USD)"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold", size = 10),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

```

Club Trajectories Through the Pandemic (2013–2024)
Predicted Log Market Value (Refitted Model)



Q2: Biological Career Arc

How does market value change as a player moves through their biological career arc (Prospect → Prime → Veteran)?

- Below we visualize the biological constraints on valuation. The curve rises steeply as players transition from ‘Prospects’ (Age 17–23), flattens into a ‘Prime’ plateau between ages 26 and 28, and declines steadily thereafter. The model predicts that a player’s value drops by approximately 1.5 log points (from a peak of 17.0 to 15.5) as they age from 27 to 35, reflecting the market’s heavy discount on older assets with limited resale potential.

```
# 1. Get Mean Age to reverse the centering
# We need the exact mean used in the dataset
mean_age <- mean(clean_player_df$Age, na.rm = TRUE)

# 2. Generate Predictions
pred_age <- ggpredict(final_model, terms = "age_c [all]")

# 3. Back-Transform X-axis
plot_data_age <- pred_age %>%
  as.data.frame() %>%
  mutate(
    # Formula: Real_Age = Centered_Age + Mean_Age
    age_real = x + mean_age
  )

# 4. Plot with Real Ages
ggplot(plot_data_age, aes(x = age_real, y = predicted)) +

  # Trend Line and Ribbon
  geom_line(color = "#3b528b", linewidth = 1.5) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
    fill = "#3b528b", alpha = 0.2) +

  # Add Peak Reference Line (Visual aid)
  geom_vline(xintercept = 27, linetype = "dashed", color = "gray50") +
  annotate("text", x = 27.5, y = min(plot_data_age$predicted) + 1,
    label = "Peak ~27", hjust = 0, color = "gray40", size = 3.5) +

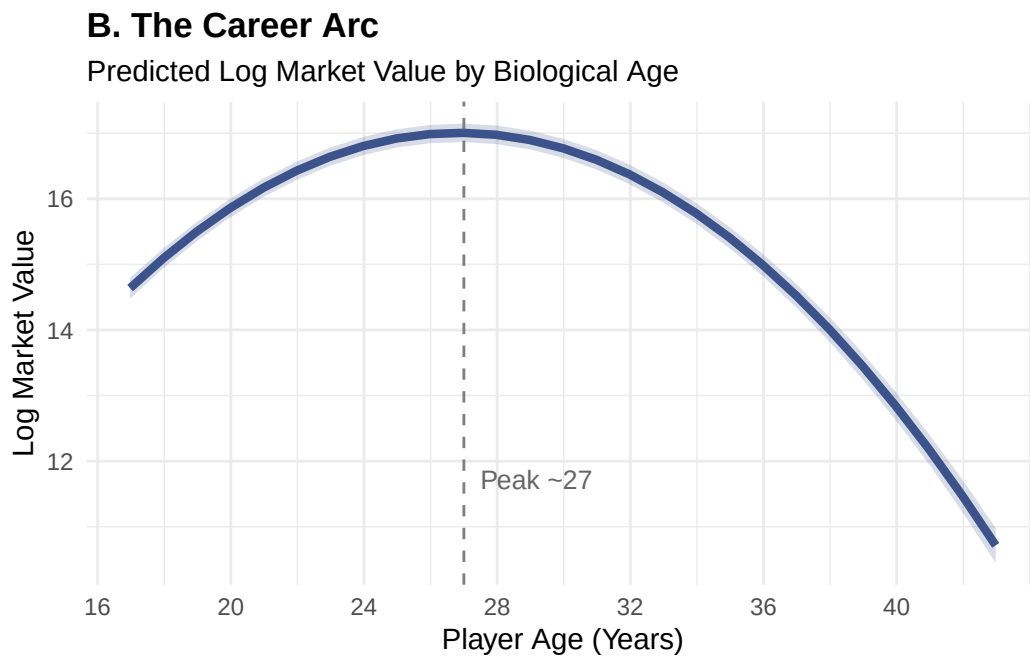
  # formatting
  scale_x_continuous(breaks = seq(16, 40, 4)) + # Show ages 16, 20, 24...

  labs(
```

```

title = "B. The Career Arc",
subtitle = "Predicted Log Market Value by Biological Age",
x = "Player Age (Years)",
y = "Log Market Value"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold"))

```



Q3: Squad Status

Does being a key player (playing more than the average teammate) increase market value?

- Figure 1C visualizes the massive premium placed on utilization. The steep linear slope confirms that 'Squad Status' is a primary driver of valuation. Moving from a rotational role (−1 SD) to a key starter role (+1 SD) is associated with an approximate doubling of market value (+101%), indicating that the market heavily discounts latent talent that cannot crack the starting XI.

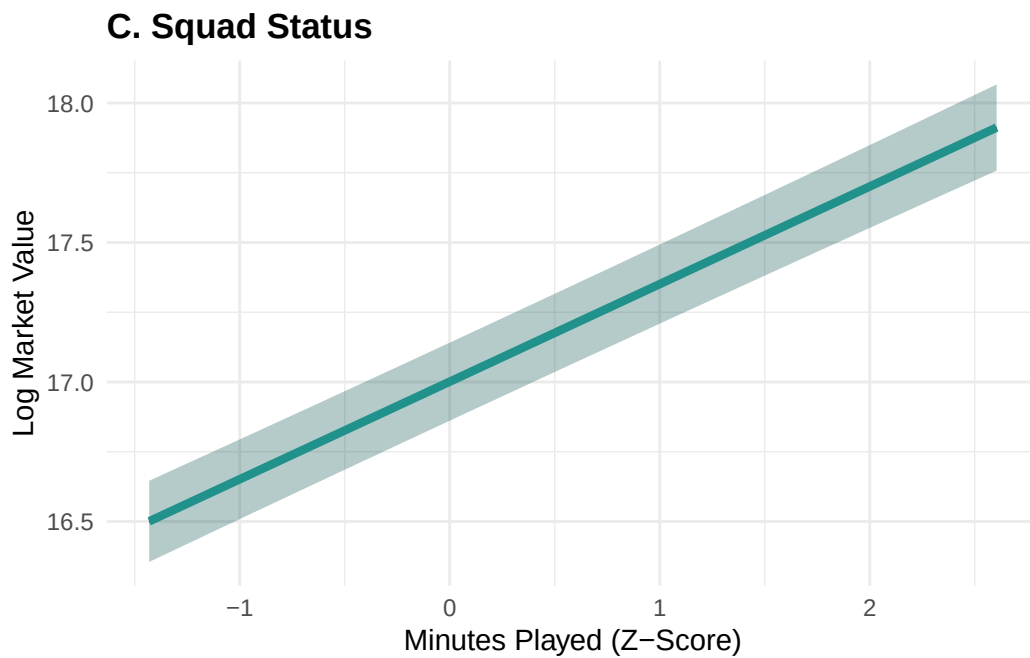
```

# Generate predictions for Minutes Played
pred_status <- ggpredict(final_model, terms = "minutes_played [all]")

plot(pred_status) +

```

```
geom_line(color = "#21918c", linewidth = 1.5) +
geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "#21918c", alpha = 0.2) +
labs(
  title = "C. Squad Status",
  x = "Minutes Played (Z-Score)",
  y = "Log Market Value"
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold"))
```



Contextual Premiums and Discounts (Q5-Q7) Figure 1D visualizes the valuation gaps across citizenship, position, and region.

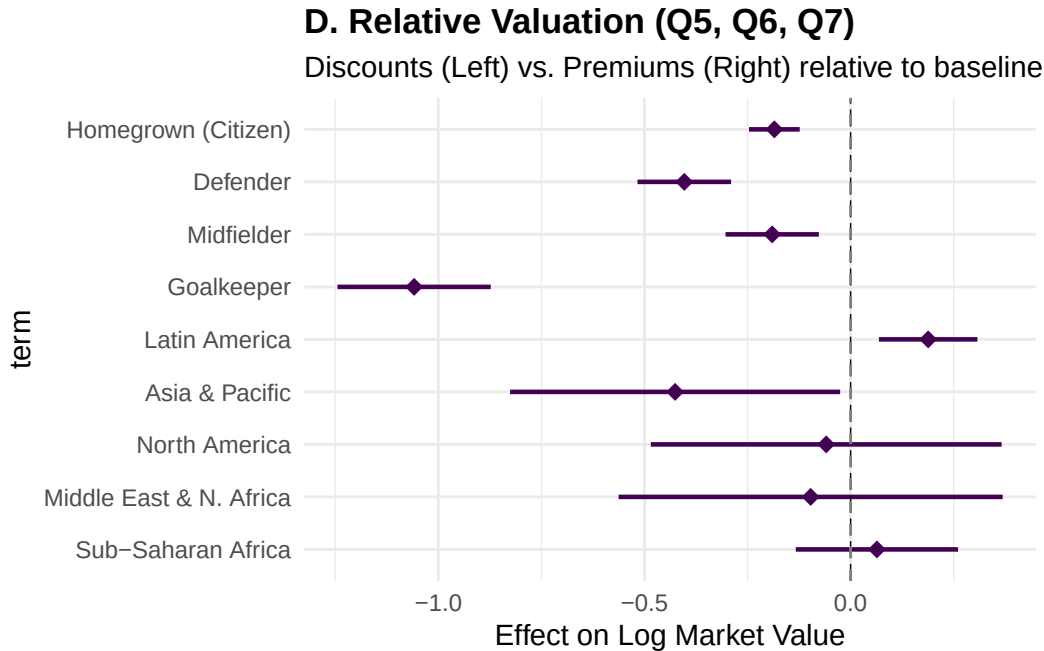
- **The “Domestic Discount” (Q5):** The coefficient for “Homegrown (Citizen)” falls strictly to the left of zero, confirming that domestic players are valued significantly lower than comparable foreign imports ($\beta = -0.19$), potentially reflecting the recruitment premium required to attract international talent.

- **Positional Hierarchy (Q6):** Attackers (the baseline) command the highest market value. All other positions show significant negative effects, with Goalkeepers facing the steepest penalty ($\beta = -1.06$), suggesting the market disproportionately rewards offensive output.
- **The “Export Brand” Effect (Q7):** While most regional coefficients overlap with zero (indicating no difference from the European baseline), **Latin America** stands out as the only region with a positive premium ($\beta = 0.19$). This suggests that South American players function as a “premium export brand” in the global market, distinct from other non-European origins.

```
plot_summs(final_model,
  coefs = c("Homegrown (Citizen)" = "citizen_of_club_countryYes",
    "Defender" = "positionDefender",
    "Midfielder" = "positionMidfield",
    "Goalkeeper" = "positionGoalkeeper",
    "Latin America" = "RegionLatin America & Caribbean",
    "Asia & Pacific" = "RegionEast Asia & Pacific",
    "North America" = "RegionNorth America",
    "Middle East & N. Africa" = "RegionMiddle East & North Africa",
    "Sub-Saharan Africa" = "RegionSub-Saharan Africa"),
  colors = "#440154", # Viridis Purple
  point.shape = 18) +

labs(title = "D. Relative Valuation (Q5, Q6, Q7)",
  subtitle = "Discounts (Left) vs. Premiums (Right) relative to baseline",
  x = "Effect on Log Market Value") +

geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
theme_minimal() +
theme(plot.title = element_text(face = "bold"))
```



Linear Trend Deviation (Center Panel): Clubs to the right (positive values) experienced faster overall value growth across the decade compared to the market average. Curvature Deviation (Right Panel): Positive values indicate a ‘Convex’ shape (accelerating growth or strong recovery), while negative values indicate a ‘Concave’ shape (diminishing returns or plateauing).

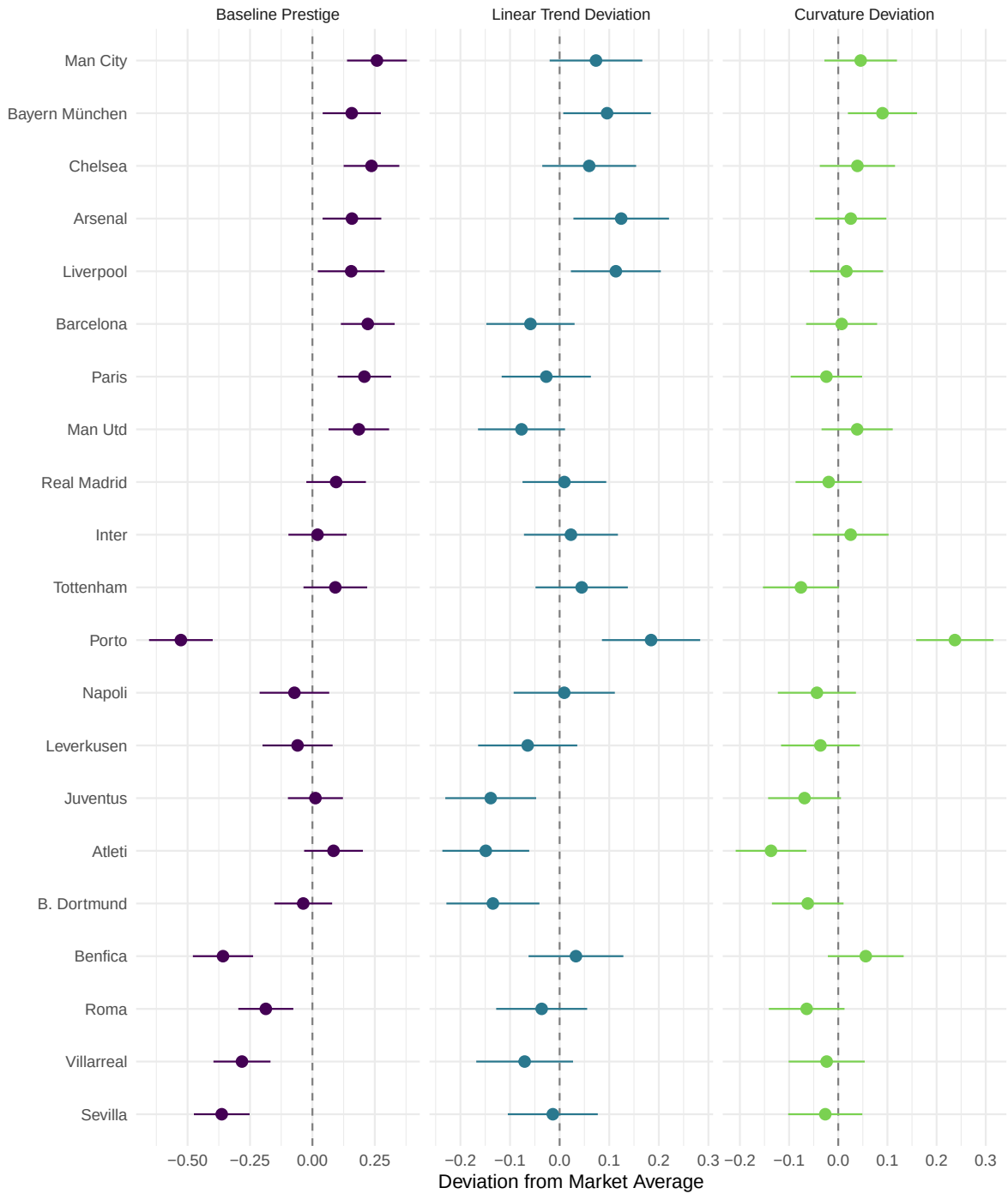
Club-Level Heterogeneity

Figure below displays the estimated club random effects from our cross-classified model. Substantial heterogeneity exists in baseline valuations, temporal trends, and curvature.

- **Baseline Prestige (Panel A):** As expected, elite ‘Super Clubs’ such as **Manchester City** and **Bayern Munich** show massive positive intercepts, indicating that their players command a premium of ~30% purely due to the club’s brand.
- **Linear Trend Deviation (Panel B):** Clubs to the right (positive values) experienced **faster overall value growth** across the decade compared to the market average. For example, **Arsenal** and **Liverpool** show strong positive deviations, confirming their role as high-growth environments.
- **Curvature Deviation (Panel C):** This panel reveals the *shape* of the trajectory. Positive values indicate a ‘Convex’ shape (accelerating growth or resilient recovery), while

negative values indicate a ‘**Concave**’ shape (diminishing returns or plateauing), highlighting the different ways clubs navigated the market cycle.

Club-Level Trajectory Deviations
Estimates from Model 3.2 (Orthogonal Polynomials)



The “Prodigy Premium”: Pricing Future Potential

The largest positive random effects belong overwhelmingly to teenage phenoms, confirming that the market prices **potential** far higher than realized performance. **Lamine Yamal** exhibits the most extreme effect in the dataset: controlling for his age, position, and playing time, his market value is approximately **19.0 times higher** (95% CI : [8.4, 43.1]) than the model predicts. Similar “super-premiums” characterize other young stars like **Gavi** (16.1×), **Florian Wirtz** (15.5×), and **Jude Bellingham** (11.8×).

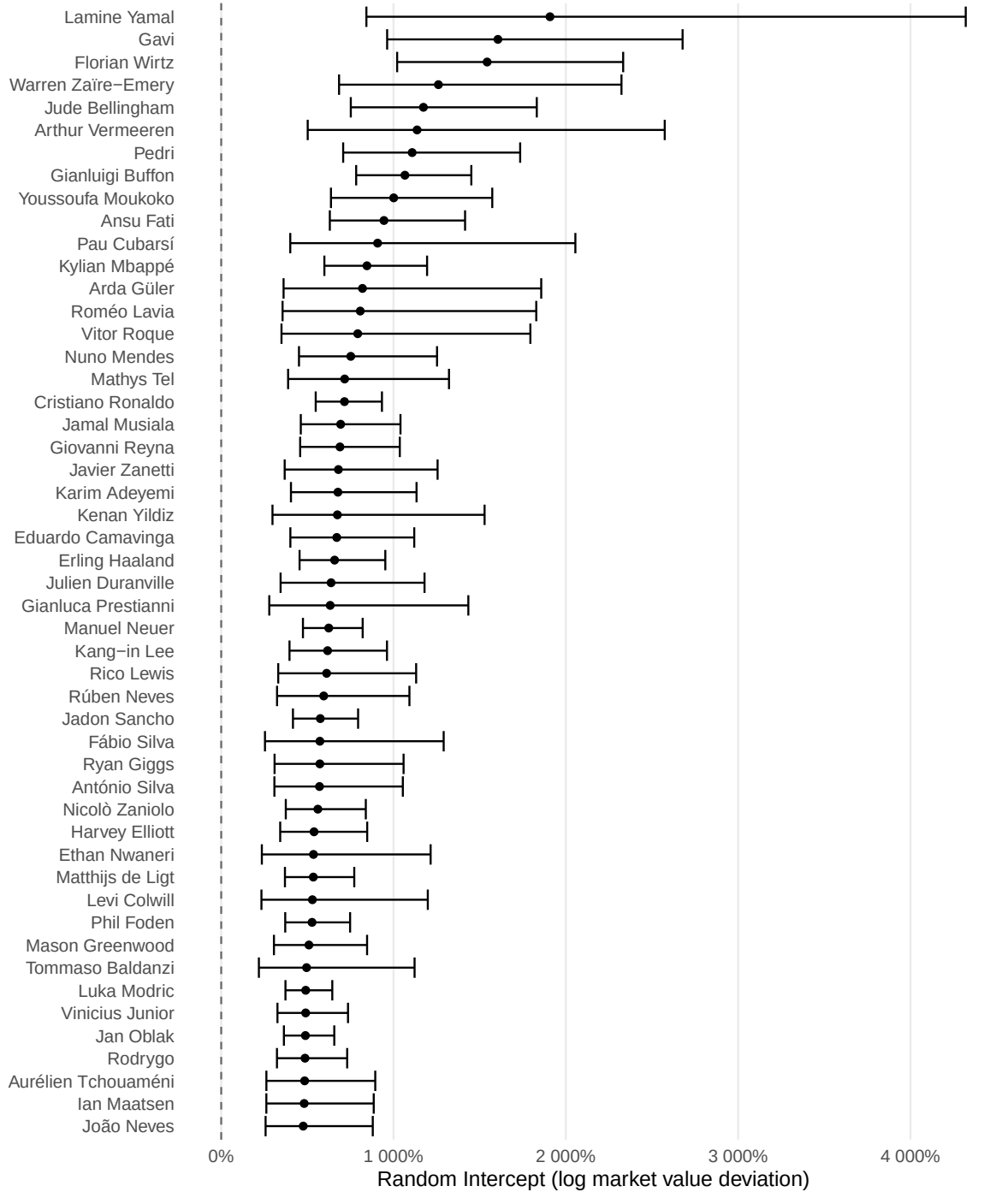
These extreme deviations highlight a fundamental feature of football valuation: **Option Value**. Clubs pay premiums for young players not for their current statistical output (which is often modest compared to prime-age veterans), but because they are purchasing a “call option” on future superstar performance. The random intercept absorbs this unmeasured “star quality”—a combination of technical ceiling and marketability that scouts perceive but standard metrics cannot capture.

The Barcelona Effect There is a striking concentration of talent from FC Barcelona among the top outliers. Five of the top 11 premiums belong to Barcelona academy graduates or players: **Lamine Yamal** (1st), **Gavi** (2nd), **Pedri** (7th, 11.1×), **Ansu Fati** (10th, 9.5×), and **Pau Cubarsí** (11th, 9.0×). This suggests a potential **institutional multiplier**: the “La Masia” brand (Barcelona’s academy) may systematically inflate the market’s valuation of its graduates, or perhaps the club provides a unique platform that accelerates reputation faster than sporting output alone.

The “Legacy Anomaly”

Interestingly, the list of outliers is not exclusively young. **Gianluigi Buffon** (8th, 10.7×) and **Cristiano Ronaldo** (18th, 7.2×) appear alongside the teenagers. This reflects a “**Legacy Premium**.” Just as prodigies are priced on future potential, icons are priced on historical achievement and commercial gravity. Our model, which relies on current-season statistics and penalizes advanced age via the quadratic term, underestimates these players because it cannot see their career history or commercial value. The random intercept thus serves as a catch-all for **Reputation**, capturing both the promise of the future (Yamal) and the glory of the past (Ronaldo).

Player-Level Market Value Premium, Top 50



Discussion

Limitations

While our model demonstrates strong predictive power ($R^2 \approx 0.89$), several limitations warrant consideration:

1. **Small Cluster Size:** We estimated a complex random effects structure (Intercept + Linear + Quadratic) for `Club`, despite our dataset containing only 21 unique clubs. While the Likelihood Ratio Test supported the inclusion of these slopes ($p < .001$), variance estimates derived from such a small number of clusters should be interpreted with caution.
2. **Normality of Residuals:** Diagnostic plots indicated deviation from normality in the extreme tails of the distribution. While the large sample size ($N = 7,023$) ensures robust estimation of fixed effects, future iterations could employ robust standard errors or a Gamma GLMM to better handle these extremes.
3. **Endogeneity:** Predictors such as `minutes_played` are likely endogenous—since better players naturally accrue more playing time—and thus should be interpreted as associations reflecting “Squad Status” rather than purely causal effects.

6.4 Conclusion

Ultimately, this research highlights the necessity of moving beyond static, single-level analyses to capture the true complexity of the football transfer market. By rigorously modeling the cross-classified intersection of individual player trajectories and dynamic club contexts, we identified a “**Great Pause**” in 2020 where market growth hit zero, a strict biological valuation peak at age 27, and a massive 42% premium for starters. As the industry continues to professionalize, frameworks that account for the interplay of human capital, institutional prestige, and market cycles will remain indispensable for distinguishing genuine value from market noise.

	Model_2.5	Model_3.2
(Intercept)	15.879 ***	17.037 ***
	(0.072)	(0.071)
poly(season__year__c, 3, raw = TRUE)1	-0.070	0.002
	(0.043)	(0.030)
poly(season__year__c, 3, raw = TRUE)2	-0.167 ***	-0.270 ***
	(0.034)	(0.024)
poly(season__year__c, 3, raw = TRUE)3	-0.084 ***	-0.117 ***
	(0.016)	(0.011)
height__c		0.010 **
		(0.004)
positionDefender		-0.403 ***
		(0.058)
positionGoalkeeper		-1.059 ***
		(0.095)
positionMidfield		-0.190 **
		(0.058)
citizen__of__club__countryYes		-0.185 ***
		(0.031)
RegionEast Asia & Pacific		-0.426 *
		(0.204)
RegionLatin America & Caribbean		0.188 **
		(0.061)
RegionMiddle East & North Africa		-0.097
		(0.238)
RegionNorth America		-0.059
		(0.217)
RegionSub-Saharan Africa		0.064
		(0.100)
poly(age__c, 2, raw = TRUE)1		0.019 ***
		(0.004)
poly(age__c, 2, raw = TRUE)2		-0.024 ***